

Recent Advancements and Future Trends in AI

World models

The technical definition of a world model is a learned function to take a state of the world and an action and predict what comes next (Huang, 2026). World models understand physics, causality, materials, objects behavior, and how the real-world works. They take different types of input (like video, audio, image, sound, movement, etc) to predict what happens next. The difference between an LLM and a World model is that the LLM learns that at certain temperature the water changes its properties and becomes ice because it had read this million times. On the other hand, a World model that has never seen this property of water (become ice at 0 Celsius degrees) would anyways predict this, since it really understands how the real-world works.

World models are important because real world data is hard to obtain and to scale. Some physical applications of AI like Autonomous Vehicles need synthetic data to train; these models would generate data based on text and visual inputs to generate new scenarios (*What Is a World Model?* | NVIDIA Glossary, n.d.).

Continual Learning

An Artificial Intelligence approach that involves training a model for new tasks while preserving previously learned tasks (*Recent Advancements and Future Trends in AI.docx*, n.d.). The problem with current algorithms is that when they are trained on new tasks the model updates its weights to complete the new task, some of those updates move the weights that are critical to previous tasks, degrading its performance (*Continual Learning and Catastrophic Forgetting Prevention in AI Agents* | Zylos Research, 2026).

Inference Time Scaling

Consists of, instead of spending time on training the model, researchers spend more time and compute power in inference time. The reasoning is that letting the models think longer with a more complex chain of thoughts produces capabilities that the models cannot achieve with regular training. The problem with this advancement is that it needs a lot of compute power, and costs three times more money than a regular model (Crosley, 2025).

EU vs Australia

The European Union's regulation on AI consists of classifying AI systems in four categories depending on their level of risk, applying huge penalizations to enterprises. High risk systems must be completely transparent, and humans must track and modify or annulate all the decisions made by the system. On the other hand, Australia is going to rely its regulation on AI on the regular laws made for all sectors, complemented by guidelines.

The European Union's regulation on AI would affect the recent advancements discussed above in a negative way. For example, since the EU's regulation requires classification of the AI system, and a complete documentation of how it works (how it was trained, what data, its parameters, etc.) a continual learning model would have problems to be updated in the European Union, since its parameters would grow or change over time to learn new tasks, making almost impossible to update the system in the EU as fast as how the system would learn. On the other hand, with the flexibility showed by Australia's regulations, the model would not need to be registered so often, making the implementation of the system easier.

In conclusion, the EU's regulations are very strict, making the implementation of new advancements very difficult for the companies. And the flexibility of Australia does the opposite, making implementation easier.

Citations:

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